

Mode

Channel4	Disable
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User Constants

 Search (Ctrl+F)

Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)

I2C2_SDA

Software Packs

Pinout

NVIC Mode and Configuration

Configuration

NVIC Code generation

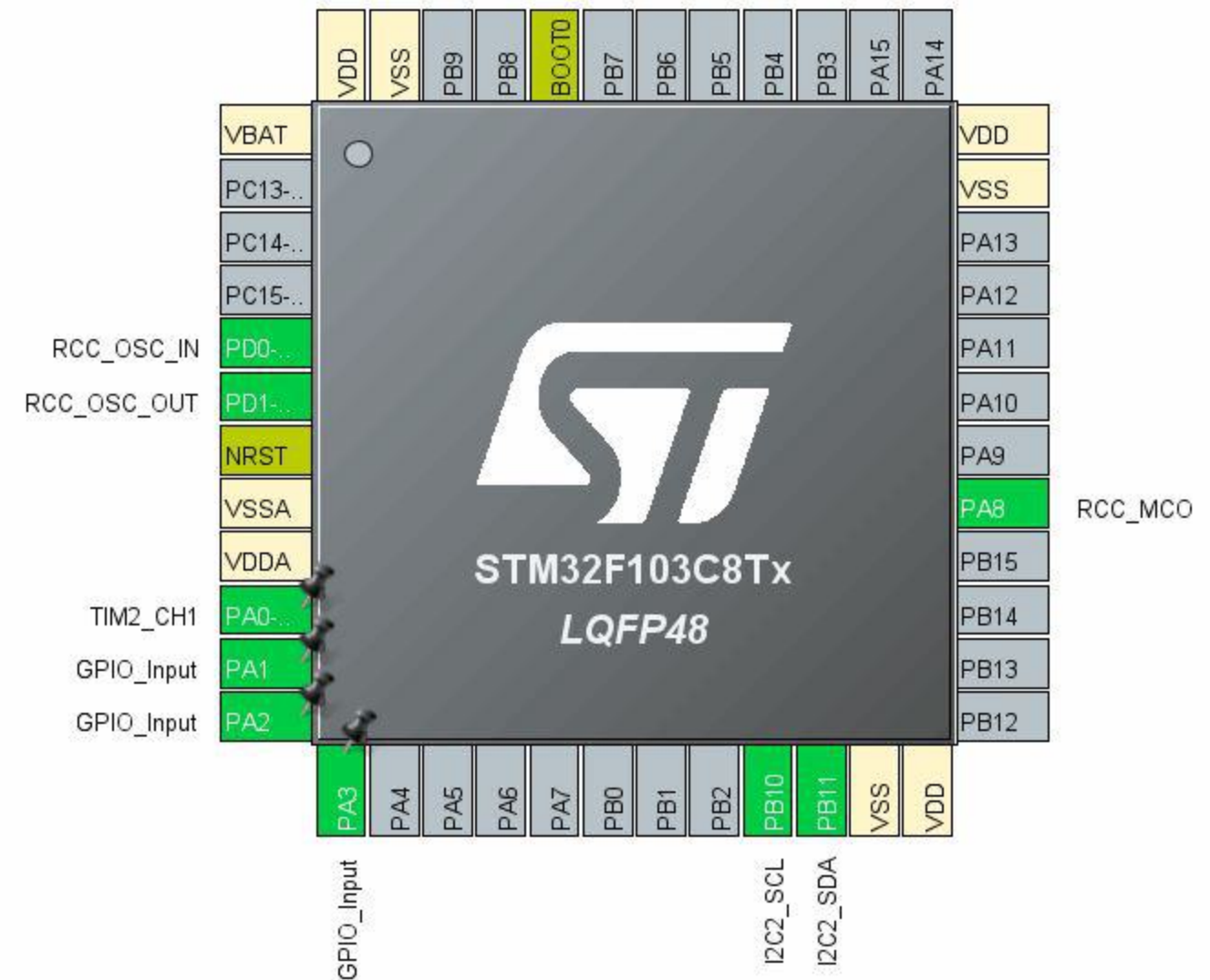
Priority Group ☐ Sort by Preemption Priority and Sub Priority ☐ Sort by interrupts nameSearch Show available interrupts ☒ Force DMA channels I

NVIC Interrupt Table	Enabled	Preemption Priority	Sub
Non maskable interrupt	☒	0	0
Hard fault interrupt	☒	0	0
Memory management fault	☒	0	0
Prefetch fault, memory access fault	☒	0	0
Undefined instruction or illegal state	☒	0	0
System service call via SWI instruction	☒	0	0
Debug monitor	☒	0	0
Pendable request for system service	☒	0	0
Time base: System tick timer	☒	15	0
PVD interrupt through EXTI line 16	☐	0	0
Flash global interrupt	☐	0	0
RCC global interrupt	☐	0	0
TIM2 global interrupt	☒	0	0
TIM3 global interrupt	☒	0	0
I2C2 event interrupt	☐	0	0
I2C2 error interrupt	☐	0	0

☐ Enabled Preemption Priority Sub Priority

Pinout view

System view



TIM3 Mode and Configuration

Mode

Slave Mode

Disable

Trigger Source

Disable

☒ Internal Clock

Channel1

Disable

Channel2

Disable

Channel3

Disable

Channel4

Disable

Configuration

Reset Configuration

☒ Parameter Settings

☒ User Constants

☒ NVIC Settings

☒ DMA Settings

NVIC Interrupt Table	Enabled	Preemption Priority	Sub Priority
TIM3 global interrupt	☒	0	0

Pinout view

System view

VDD

VSS

PB9

PB8

BOOT0

PB7

PB6

PB5

PB4

PB3

PA15

PA14

VBAT

PC13-...

PC14-...

PC15-...

PD0-...

PD1-...

NRST

VSSA

VDDA

PA0-...

PA1

PA2

PA3

PA4

PA5

PA6

PA7

PB0

PB1

PB2

PB10

PB11

VSS

VDD

PA13

PA12

PA11

PA10

PA9

PA8

PB15

PB14

PB13

PB12

STM32F103C8Tx

LQFP48

RCC_OSC_IN

RCC_OSC_OUT

GPIO_Output

GPIO_Input

GPIO_Input

RCC_MCO

_Input

_SCL

_SDA

TIM3 Mode and Configuration

Mode

Slave Mode

Disable

Trigger Source

Disable

☒ Internal Clock

Channel1

Disable

Channel2

Disable

Channel3

Disable

Channel4

Disable

Configuration

Reset Configuration

Parameter Settings

User Constants

NVIC Settings

DMA Settings

Configure the below parameters :

▼ Counter Settings

Prescaler (PSC - 16 bits value)

7199

Counter Mode

Up

Counter Period (AutoReload Regist...

4999

Internal Clock Division (CKD)

No Division

auto-reload preload

Disable

▼ Trigger Output (TRGO) Parameters

Master/Slave Mode (MSM bit)

Disable (Trigger input effect not delayed)

Trigger Event Selection

Reset (UG bit from TIMx_EGR)

Pinout view

System view

RCC_OSC_IN

RCC_OSC_OUT

TIM2_CH1

GPIO_Input

GPIO_Input

VDD

VSS

PB9

PB8

BOOT0

PB7

PB6

PB5

PB4

PB3

PA15

PA14

VBAT

PC13-...

PC14-...

PC15-...

PD0-...

PD1-...

NRST

VSSA

VDDA

PA0-...

PA1

PA2

PA3

PA4

PA5

PA6

PA7

PB0

PB1

PB2

PB10

PB11

VSS

VDD

STM32F103C8Tx

LQFP48

PA8

PB15

PB14

PB13

PB12

PA13

PA12

PA11

PA10

PA9

RCC_MCO

I2C_SCL

I2C_SDA

TIM3 Mode and Configuration

Mode

Slave Mode

Disable

Trigger Source

Disable

☒ Internal Clock

Channel1

Disable

Channel2

Disable

Channel3

Disable

Channel4

Disable

Configuration

Reset Configuration

☒ Parameter Settings

☒ User Constants

☒ NVIC Settings

☒ DMA Settings

NVIC Interrupt Table	Enabled	Preemption Priority	Sub Priority
TIM3 global interrupt	☒	0	0

Pinout view

System view

VDD

VSS

PB9

PB8

BOOT0

PB7

PB6

PB5

PB4

PB3

PA15

PA14

VBAT

PC13-..

PC14-..

PC15-..

PD0-..

PD1-..

NRST

VSSA

VDDA

PA0-..

PA1

PA2

PA3

PA4

PA5

PA6

PA7

PB0

PB1

PB2

PB10

PB11

VSS

VDD

GPIO_Input

I2C2_SCL

I2C2_SDA

RCC_OSC_IN

RCC_OSC_OUT

TIM2_CH1

GPIO_Input

GPIO_Input

VDD

VSS

PA13

PA12

PA11

PA10

PA9

PA8

PB15

PB14

PB13

PB12

RCC_MCO

STM32F103C8Tx

LQFP48

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```
#include "main.h"

I2C_HandleTypeDef hi2c2;
TIM_HandleTypeDef htim2;
TIM_HandleTypeDef htim3;

uint8_t target_temp = 30;
uint8_t current_temp = 0;
uint8_t system_on = 1;

void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_I2C2_Init(void);
static void MX_TIM2_Init(void);
static void MX_TIM3_Init(void);

int main(void)
{
    HAL_Init();
    SystemClock_Config();
    MX_GPIO_Init();
    MX_I2C2_Init();
    MX_TIM2_Init();
    MX_TIM3_Init();

    HAL_TIM_Base_Start_IT(&htim3);
    HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);

    while (1)
    {
        if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_1) == GPIO_PIN_RESET)
        {
            HAL_Delay(50);
            if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_1) == GPIO_PIN_RESET)
            {
                system_on = !system_on;
                while(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_1) == GPIO_PIN_RESET);
            }
        }
    }
}
```

```
if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_2) == GPIO_PIN_RESET)
{
    HAL_Delay(50);
    if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_2) == GPIO_PIN_RESET)
    {
        target_temp += 2;
        while(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_2) == GPIO_PIN_RESET);
    }
}

if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_3) == GPIO_PIN_RESET)
{
    HAL_Delay(50);
    if(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_3) == GPIO_PIN_RESET)
    {
        target_temp -= 2;
        while(HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_3) == GPIO_PIN_RESET);
    }
}
}
```

```
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef
*htim)
{
    if(htim == &htim3)
    {
        uint8_t reg_addr = 0x00;
        HAL_I2C_Master_Transmit(&hi2c2, (0x48 << 1), &reg_addr, 1,
10);
        HAL_I2C_Master_Receive(&hi2c2, (0x48 << 1), &current_temp,
1, 10);

        if(system_on == 1)
        {
            if(current_temp > target_temp)
            {
                uint32_t speed = (current_temp - target_temp) * 100;
                if(speed > 999) speed = 999;
                __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1,
speed);
            }
            else
            {
                __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 0);
            }
        }
        else
        {
            __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, 0);
        }
    }
}
```